

Business Report

DePauw Loss Minimizers NCAA Men's Final Four Analytics Challenge

I. Introduction

Each year, the NCAA Selection Committee faces a critical decision problem: determining which 68 teams qualify for the tournament and how they should be seeded. While the tournament itself captures attention through unpredictable outcomes and high-stakes competition, the selection process behind it remains complex, multi-dimensional, and often debated.

These decisions have significant implications across the NCAA ecosystem. Tournament selection influences team revenue, national exposure, recruiting opportunities, and fan engagement. Given these high stakes, the ability to understand and evaluate this process in a consistent and objective manner is essential.

In this project, we aim to replicate and improve the NCAA selection process using a data-driven approach. Specifically, we focus on two key tasks:

1. Predicting whether a team qualifies for the tournament
2. Predicting its tournament seed among selected teams

By analyzing performance metrics, résumé-based features, and strength of competition, we seek to identify the factors that most strongly influence selection outcomes. Our goal is not only to improve predictive accuracy, but also to provide insights into how selection decisions are made and how teams can improve their chances of success.

II. Data and Constraints - What We Work With

Data Sources and Enrichment

To build a reliable and realistic prediction system, we combined multiple data sources. In addition to the official NCAA dataset, we incorporated external data from KenPom (advanced analytics) and ESPN (traditional box score statistics). This added 16 additional features, including:

- Scoring: Points per game, FG%, 3P%, FT%
- Playmaking: Assists, turnovers
- Defense: Steals, blocks
- Rebounding: Offensive, defensive, total rebounds

- Adjusted offensive & defensive efficiency
- Tempo (possessions per game)
- Luck rating

Constraints and Solutions

1. Highly Imbalanced Dataset

In the training dataset, about 1,100 data points are Non-Compete teams while only about 250 of the data points are Compete teams, which is roughly 18%. This means most teams do not make the tournament and a naive model could over-predict Non-Compete teams. So, we instruct the model to weigh the minority group heavier to avoid biases.

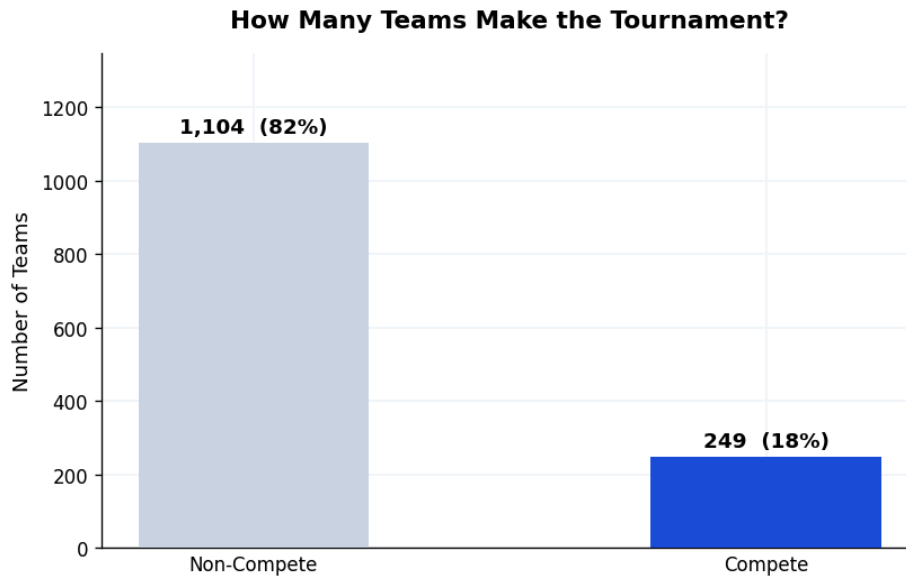


Figure 1: Count of Tournament Participants

2. No Explicit Committee Rules

The NCAA decisions are subjective and multi-factor with no official formula exists to calculate participation possibilities and Seed Ranking. Therefore, our challenge is to build a model that can learn decision patterns indirectly from data. We go ahead and construct variables that capture performance quality, opponent strength, consistency, and context.

3. Generalization Across Seasons

Every year, the NCAA has different teams participating in. This makes team strength, competition level, and rankings vary each year. To address this, we validate the model using season-based splits to ensure the model learns consistent patterns and not just memorizing specific years.

III. Key Insights - What Drives Tournament Decisions

Instead of focusing on raw features, we translate them into basketball insights that explain how teams are evaluated.

1. Quality Wins Matter More Than Total Wins

Our analysis shows that teams with strong performance against top-tier opponents are significantly more likely to qualify. This includes wins against high-quality teams (Quadrant 1 and 2 opponents) and “Elite wins” that demonstrate competitiveness at the highest level. In other words, who you beat matters more than how many you beat.

2. Avoiding Bad Losses Is Critical

We observe a strong negative relationship between losses to weaker teams and tournament qualification. Even a small number of poor losses can significantly reduce a team’s chances. This reflects real committee behavior. Consistency matters, and weak performances are heavily penalized.

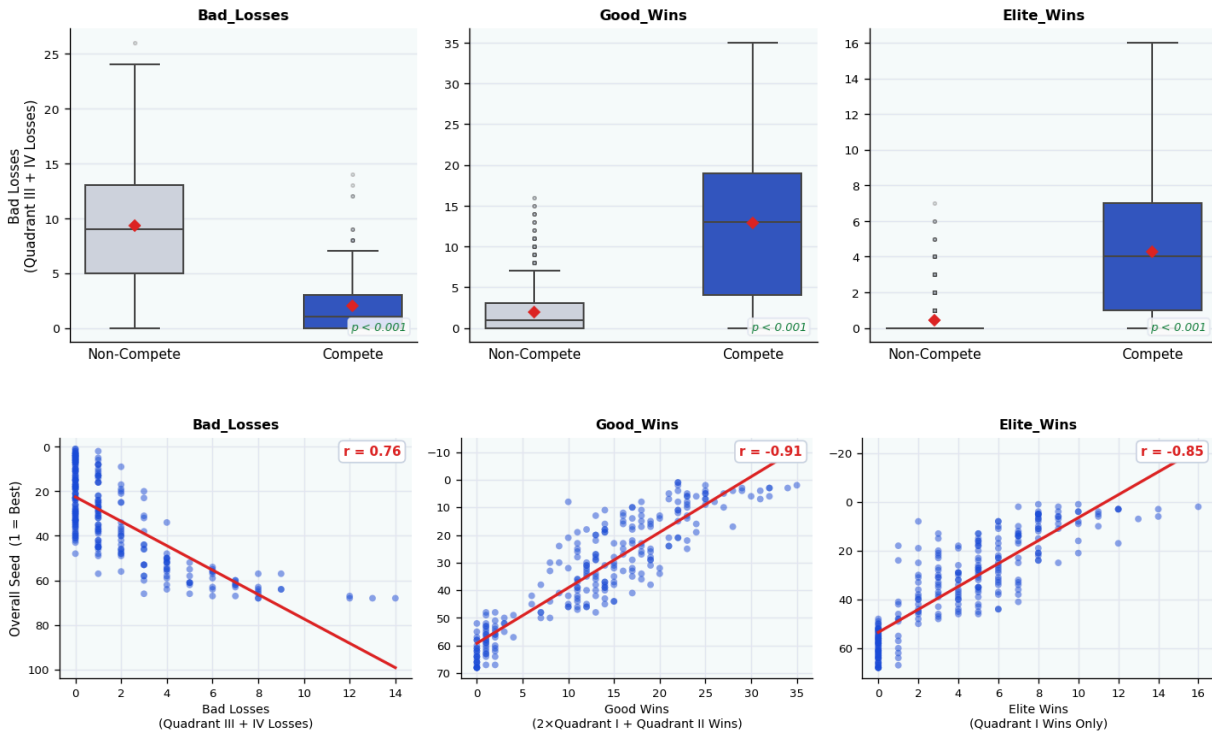


Figure 2: Important Wins and Losses on Tournament Participation and Seeding

3. Efficiency Reflects True Team Strength

Efficiency metrics, specifically the difference between offensive and defensive performance, are among the strongest indicators of success. Teams with higher efficiency margins win more consistently and perform better against strong opponents. This captures not just outcomes, but how convincingly teams perform

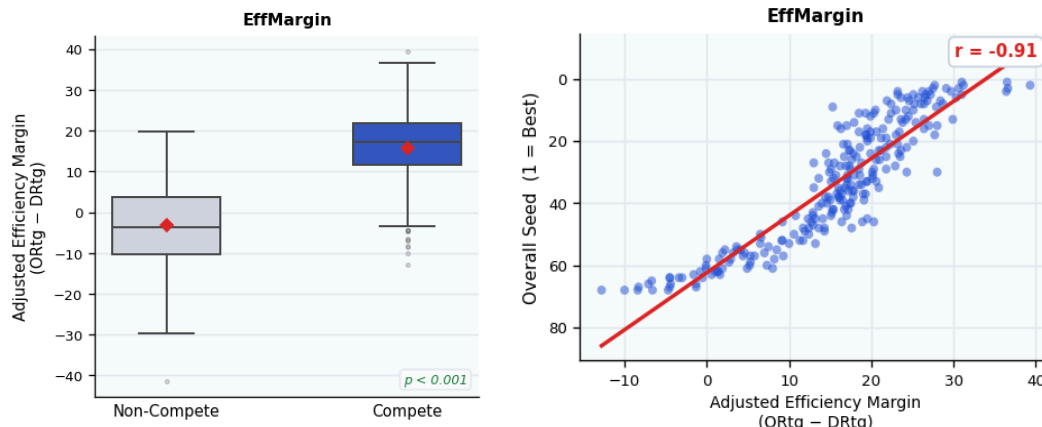


Figure 3: Efficiency Margin on Tournament Participation and Seeding

4. Exposure to Strong Competition Matters

Teams that regularly face strong opponents and compete in high-level games are evaluated differently from those with weaker schedules. Key indicators include win rate against top-tier teams, the average strength of opponents, and number of high-quality games played. This reflects the idea that strong teams are tested, not just successful.

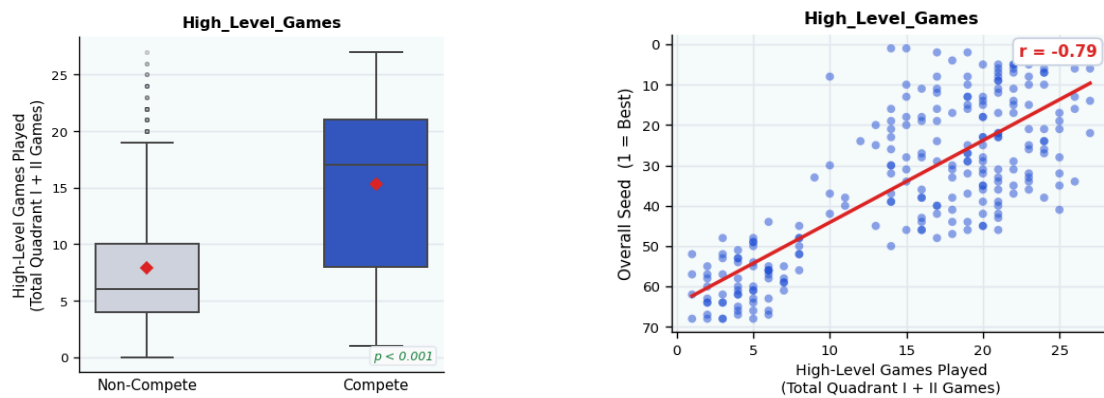


Figure 4: Strong Exposure on Tournament Participation and Seeding

5. Ranking Is Relative, Not Absolute

A key insight is that team strength must be evaluated within the context of a given season. We constructed a composite ranking system that combines performance metrics, strength of schedule, and win percentages. While individually smaller, together they provide a more complete picture of team performance. This allows us to assess how strong a team is relative to its competition in the same year.

Across all analyses, one consistent message is that tournament selection is not driven by a single metric, it is the result of balanced performance across strength, consistency, and competition level. Our model captures this by combining performance quality, opponent strength, consistency, and context.

IV. Solution - A Two-Step Decision System

To mirror how the NCAA selection committee operates, we designed a two-step predictive system that separates:

- Tournament Selection (Who gets in?)
- Tournament Seeding (How strong are they?)

This structure is intentional and critical. In reality, the NCAA does not rank all teams equally. It first selects the top 68 teams, then assigns seeds only among those teams. By replicating this workflow, not only we reduce noise from non-competitive teams, improve prediction accuracy, we also align our model with real-world decision logic.

Step 1: Tournament Selection (Classification Model)

Our objective is to predict whether a team qualifies for the NCAA tournament with a binary outcome. We use XGBoost, a tree-based ensemble model, because it:

- Captures nonlinear relationships between team performance metrics
- Handles feature interactions automatically
- Performs well on structured sports data
- Is robust to imbalanced datasets

For starters, to address the imbalance dataset, we apply class weighting (`scale_pos_weight`) within our classifier to ensure the model does not overpredict Non-Competing teams and remains sensitive to identifying true tournament teams.

When training our classifier, we focus on two key metrics:

- ROC-AUC (Primary Metric) measures how well the model separates tournament vs non-tournament teams. A higher score means that the model ranks good teams above bad teams consistently.
- F1 Score balances missing real tournament teams (false negatives) and incorrectly selecting weak teams (false positives)

Lastly, we validate stability across seasons and generalization performance using GroupKFold (by Season) to ensure realism instead of random splits. This ensures no data leakage across seasons and the model truly learns patterns that generalize to future years. This mimics real-world deployment that predicts a new season using past seasons.

After step 1, each team receives a probability of making the tournament. We then select the top 68 teams by probability to seed. The rest are marked as non-competing teams.

Step 2: Tournament Seeding (Regression Model)

Our objective is to predict the Overall Seed (1-68) for tournament teams only

We use CatBoost, a gradient boosting model optimized for ranking and tabular data, as our regressor. CatBoost has advantages in:

- Handling nonlinear relationships
- Captures complex feature interactions
- Naturally supports categorical variables
- Strong performance in ranking-style problems

To ensure interpretability and efficiency, we use SHAP (SHapley Additive Explanations). SHAP quantifies how much each feature contributes to predictions. It acts like a feature importance scoreboard. From SHAP analysis, the most influential features include:

- Opponent Strength: AvgOppNET, NETSOS
- Core Ranking Metrics: NET Rank, Previous NET Rank
- Efficiency Measures: EffMargin
- Performance Indicators: Quadrant Wins and Losses
- Derived Features: Interaction terms (e.g., $\text{NET} \times \text{Win\%}$)
- Season-relative rankings: CompositeRank

These features align closely with how the NCAA committee evaluates teams:

- Strength of schedule matters
- Beating strong opponents is critical
- Efficiency reflects true team strength
- Rankings are relative within each season

This confirms that the model is not just predictive, it is behaviorally aligned with real decision-making.

Lastly, we measure performance using Root Mean Squared Error (RMSE) that captures how close predicted seeds are to actual seeds and penalizes large errors more heavily. This means lower RMSE shows more accurate seed predictions.

V. Results & business Impact

1. Model Performance

a. Tournament Selection (Classification)

XGBoost was selected as our final classification model after a full grid search across hundreds of hyperparameter combinations, all validated using GroupKFold by season. The selected model achieved a test ROC-AUC of 98.01% and a test F-1 score of 83.6%. The gap between average ROC-AUC training and testing set was just 1.95%.

ROC-AUC score measures how well the model separates tournament teams from non-tournament teams. A 98% ROC-AUC means that if we randomly pick one tournament team and one non-tournament team, the model will correctly rank the tournament team higher 98% of the time. A gap of 1.95% is very small compared to 98%, meaning the model performs almost equally well on unseen seasons as it does on training data. This shows that the model has committee-level decision quality in identifying which teams belong in the tournament.

F-1 score balances precision (emphasizes False Positives) and Recall (emphasizes False Negatives). An F1 of 83.6% means the model does a strong job both identifying real tournament teams and avoiding false selections. We are not just accurate, we are selecting the right teams consistently. This avoids missing strong teams (false negatives) and over-selecting weak teams (false positives).

b. Tournament Seeding (Regression)

CatBoost was selected as the regression model to predict seeds among qualified teams. Under GroupKFold validation by season, it produced a mean validation RMSE of 5.33 with a moderated gap of 2.48 between training and testing sets.

RMSE measures how far predictions are from actual seeds. On average, predicted seeds fall within 5 to 6 positions away from the true committee assignment, a meaningful result given that seeding involves no public formula and considerable subjective judgment. For example, if a team is actually seeded at 4, the models typically predict between 1 and 10. The model provides highly competitive ranking accuracy that is good enough for bracket predictions, team evaluation, and media forecasting.

2. Deployment

The most practical application of this system is a web-based dashboard where users input a team's current season statistics and receive two outputs: probability of making the tournament and predicted seed. This would serve coaches, analysts, and conference administrators as a live decision support tool throughout the season.

On the backend, the pipeline would ingest new season data each offseason, retrain models automatically, and re-evaluate performance before the next cycle. With further development, the system could connect directly to live data feeds from KenPom and ESPN, enabling real-time updates as games are played and eliminating manual data entry entirely.

3. Business Impact

a. Athletic Programs (Coaches and ADs)

Currently, most athletic departments make high-stakes scheduling decisions based on anecdotal evidence rather than quantified projections. This model empowers Coaches and Athletic Directors to understand precisely how individual matchups influence their NET rank, Strength of Schedule (SOS), and Quadrant 1 record. By identifying "Bubble Risk" 3–5 weeks before Selection Sunday, the tool provides a critical window for programs to adjust their late-season strategy or non-conference scheduling while they still have the agency to influence their tournament trajectory.

b. Conference Offices

Conference offices often suffer from a lack of coordinated strategy, with individual programs optimizing their own resumes at the expense of the collective group. This model provides administrators with a unified, forward-looking view of how every program's metrics are evolving in real-time. By coordinating schedules to secure just one or two additional at-large bids, a conference can generate an estimated additional annual NCAA distributions. This shifts the conference's role from passive observation to active portfolio management.

c. Media and Analysts

Unlike traditional bracketology, which often relies on subjective "expert" intuition and shifting methodologies, this framework utilizes a dual-stage analytical pipeline that mirrors the Selection Committee's actual deliberation logic. By separating at-large selection from the seeding process, the model generates consistent, auditable predictions that eliminate the inconsistencies of human bias. When the model's outputs diverge from the media consensus, the system isolates the specific metric shifts driving the variance, providing analysts with a defensible, data-driven foundation for their projections that significantly enhances both transparency and editorial credibility.

d. The NCAA

As tournament selection faces increasing public and legal scrutiny, the need for a transparent, documented decision-support process has never been higher. This model demonstrates that while the process is officially subjective, it follows consistent, learnable patterns. The NCAA can utilize this system as a "Consistency Audit" to flag anomalous selections before they are finalized and to simulate how proposed rule changes would impact historical data. Implementing such a benchmark ensures that high-stakes financial decisions are behaviorally aligned and institutionally prudent.

4. Future Improvements

To further increase predictive accuracy, the strategic roadmap involves shifting from season averages to Game-Level True Elo and Injury-Adjusted Personnel Logic. By tracking game-by-game momentum, the model can identify teams "peaking" at the critical juncture of the tournament. Furthermore, integrating player-level data allows the system to adjust team strength when star players are sidelined, context that standard metrics often miss. These enhancements move the project into a Live Mode forecasting tool, capable of tracking real-time performance trajectories through automated data feeds.

VI. Conclusion

Each year, the NCAA Selection Committee makes decisions that carry significant financial and reputational consequences for programs across the country, yet the process operates without a formal formula or full transparency. This project set out to answer a critical question: "Can this seemingly subjective process be translated into a consistent, data-driven system?" Our results show that it can.

The XGBoost classification model achieved a test ROC-AUC of 98%, with minimal overfitting and validation across completely unseen seasons. This demonstrates that tournament selection is not random or purely subjective. It follows consistent, learnable patterns that can be reliably applied to future seasons. More importantly, the model reveals what actually drives those decisions. Résumé-based features, such as Quadrant performance, strength of schedule, and quality wins, consistently outperform traditional box score metrics. This confirms that in tournament selection, how you win matters far more than how much you score.

The seeding model introduces a more nuanced perspective. With an RMSE of approximately 5.3 seeds, it shows that while selection is highly structured, precise ranking among top teams remains inherently more variable. Recognizing this distinction strengthens the credibility of the overall system. From a business standpoint, the implications are athletic programs can proactively improve tournament outcomes through smarter scheduling and performance targeting; conference offices can coordinate strategies to maximize bids and national visibility;

media and analysts gain a transparent, repeatable framework for explaining and predicting selections; and the NCAA itself can leverage this as a decision-support tool to improve consistency, accountability, and trust in a process tied to millions in revenue. Ultimately, what appears to be a subjective process is, in practice, a structured decision system driven by measurable factors. The committee’s decisions are not unpredictable, they are simply not yet formalized. This work brings that structure to light.

VII. Appendix

1. Training Tournament Seeds

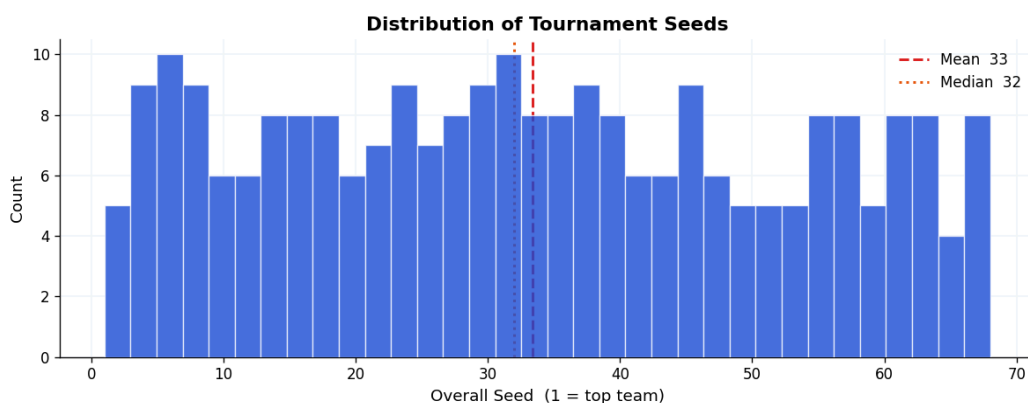


Figure 5: Distribution of Tournament Seeds

2. Win and Loss Scores

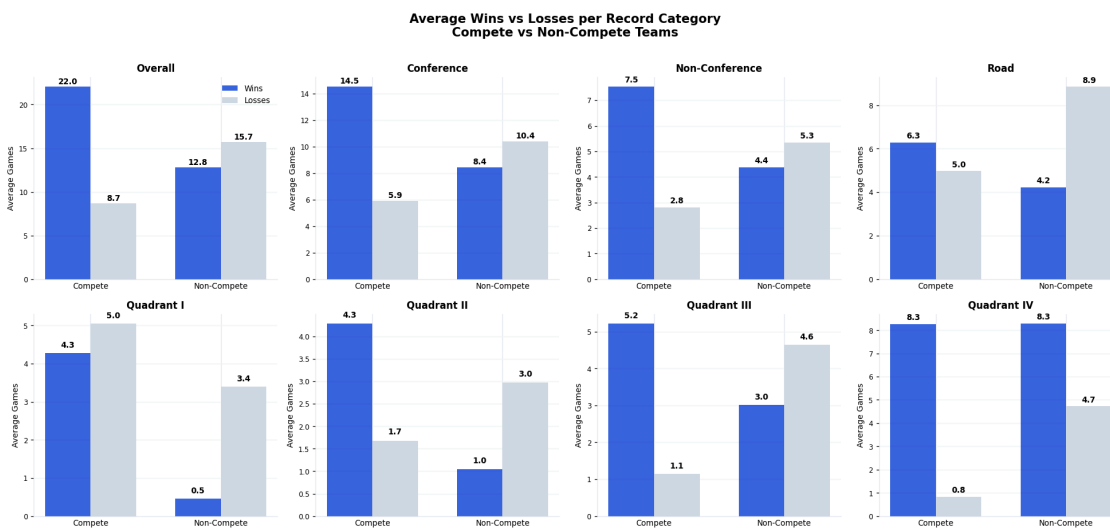


Figure 6: Count of Wins and Losses per Record Category on Tournament Participation

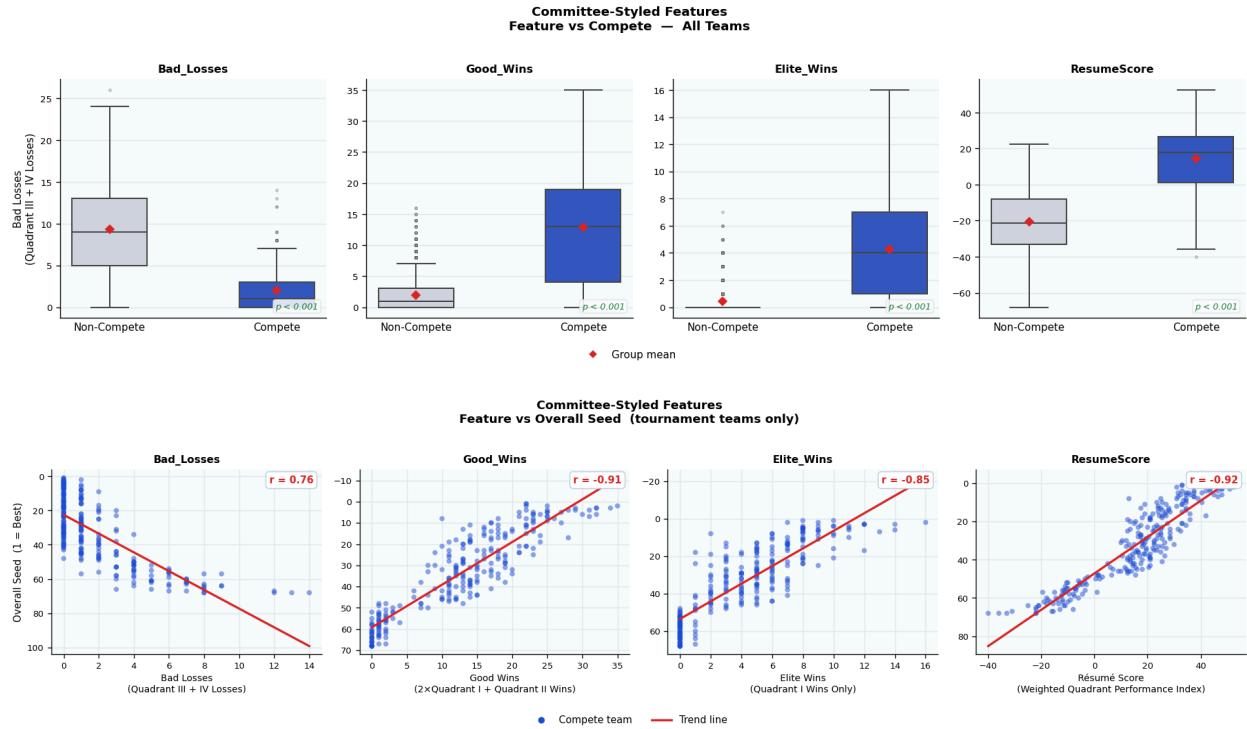


Figure 7: Average Wins and Losses on Tournament Participation and Tournament Seed

Feature Explanation:

- **Bad_Losses** count damaging losses, especially in lower quadrants.
- **Good_Wins** rewards wins over strong opponents.
- **Elite_Wins** focuses specifically on Quadrant 1 wins.
- **Resume Score** combines quality wins and damaging losses into a single résumé metric.

3. Résumé-Styled Features

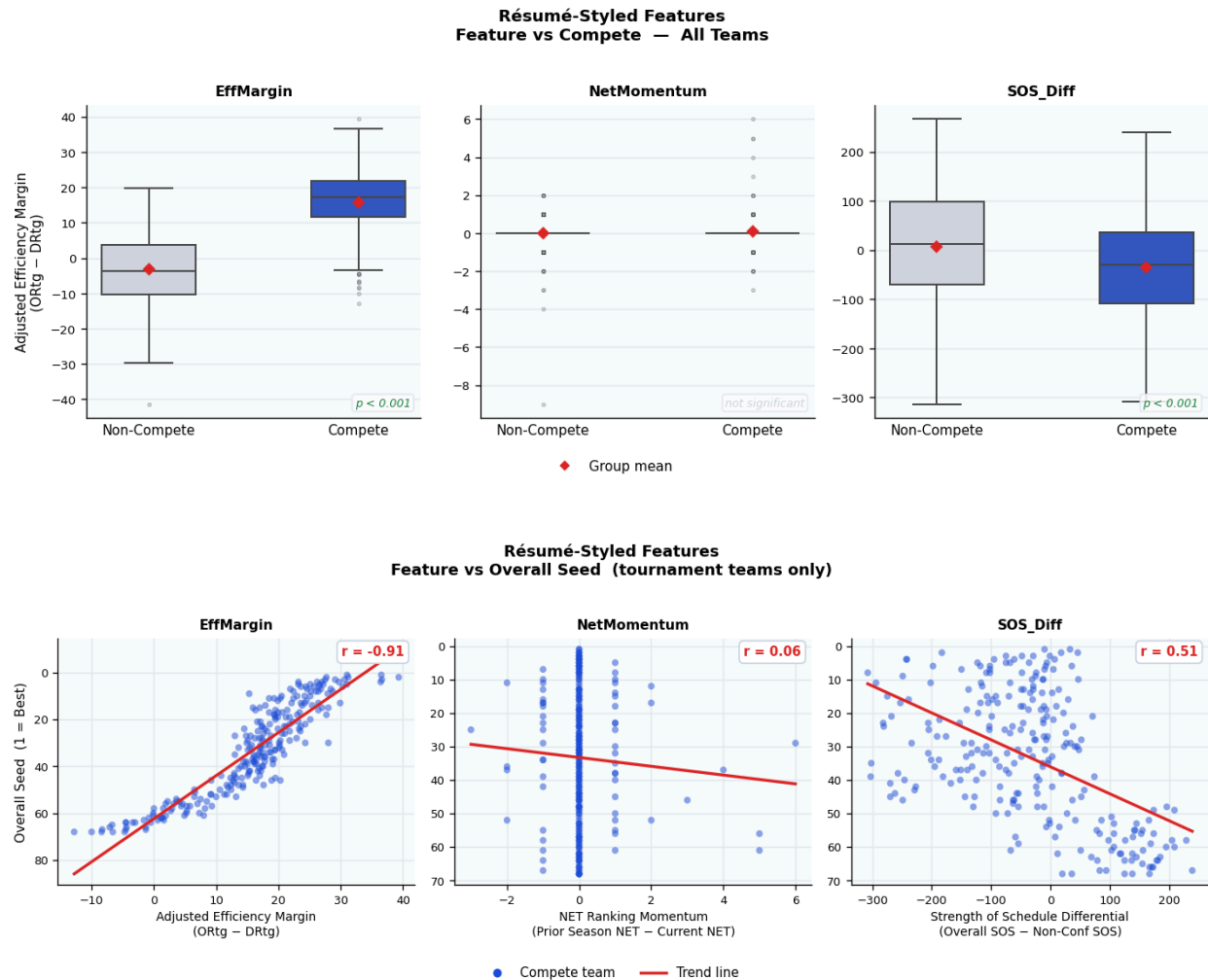


Figure 8: Average Résumé-Styled Features on Tournament Participation and Tournament Seed

Feature Explanation:

- EffMargin is offensive rating minus defensive rating. In plain English, it measures how strongly a team outperforms opponents on both ends of the floor. It is understandable that Competing Teams, on average, would have a higher EffMargin than Non-Competing Teams, as shown in the first box plot on the left.
- NetMomentum measures whether a team improved relative to the prior season or prior ranking baseline.
- SOS_Diff compares overall strength of schedule versus non-conference schedule, helping us see whether a team challenged itself consistently or mostly within conference play.

4. Strong Opponent Exposure Features

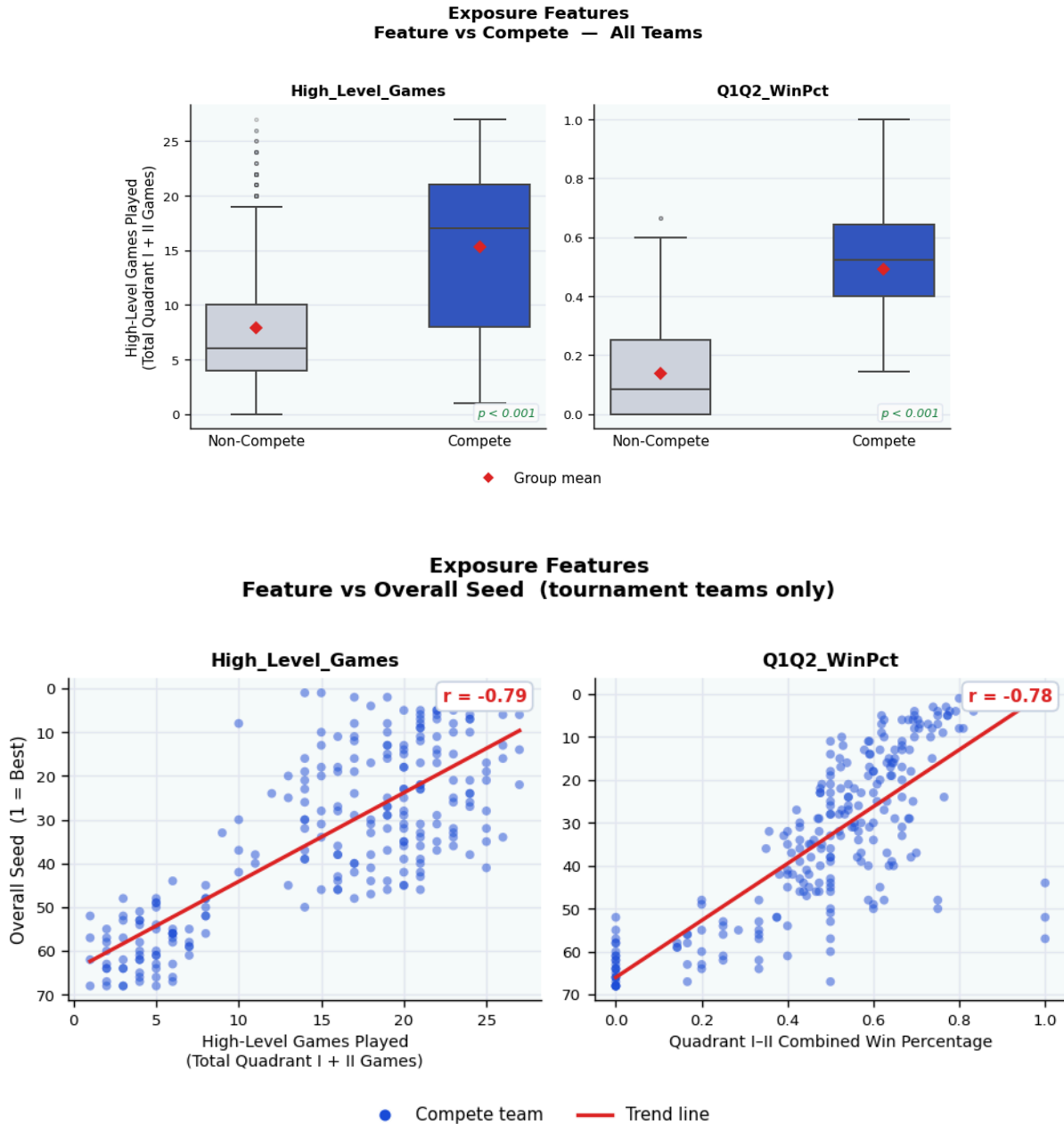


Figure 9: Average Exposure Features on Tournament Participation and Tournament Seed

Feature Explanation:

- High_Level_Games measures how often a team played strong competition - basically, how often they were tested.
- Q1Q2_WinPct measures how efficiently they performed in those high-level opportunities.

5. Ratio Features

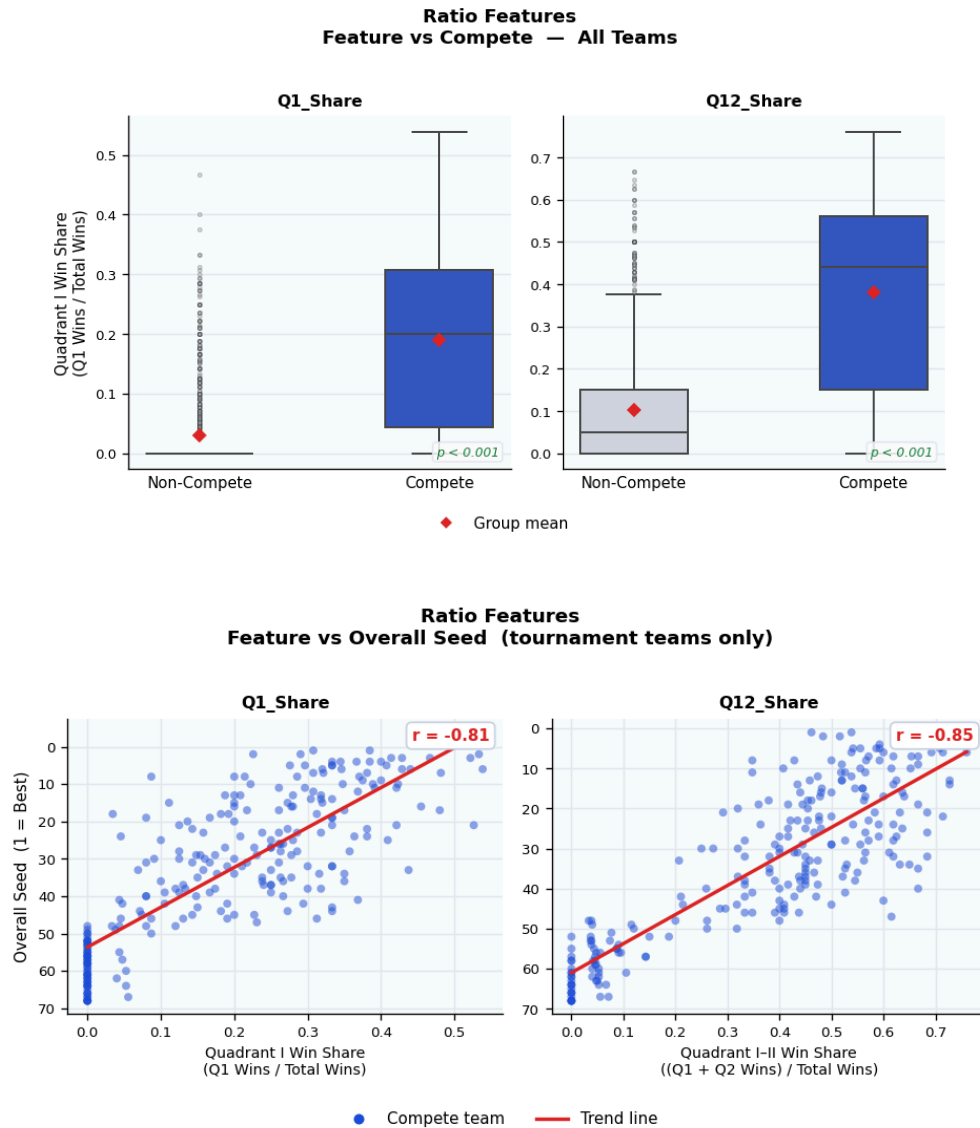


Figure 10: Average Ratio Features on Tournament Participation and Tournament Seed

Feature Explanation:

- Q1_Share and Q12_Share capture what share of a team's wins came against high-quality opponents (Quadrant 1 and 2). This prevents a team with many wins against weak teams from looking artificially strong.

6. Committee-like Interaction Terms Features

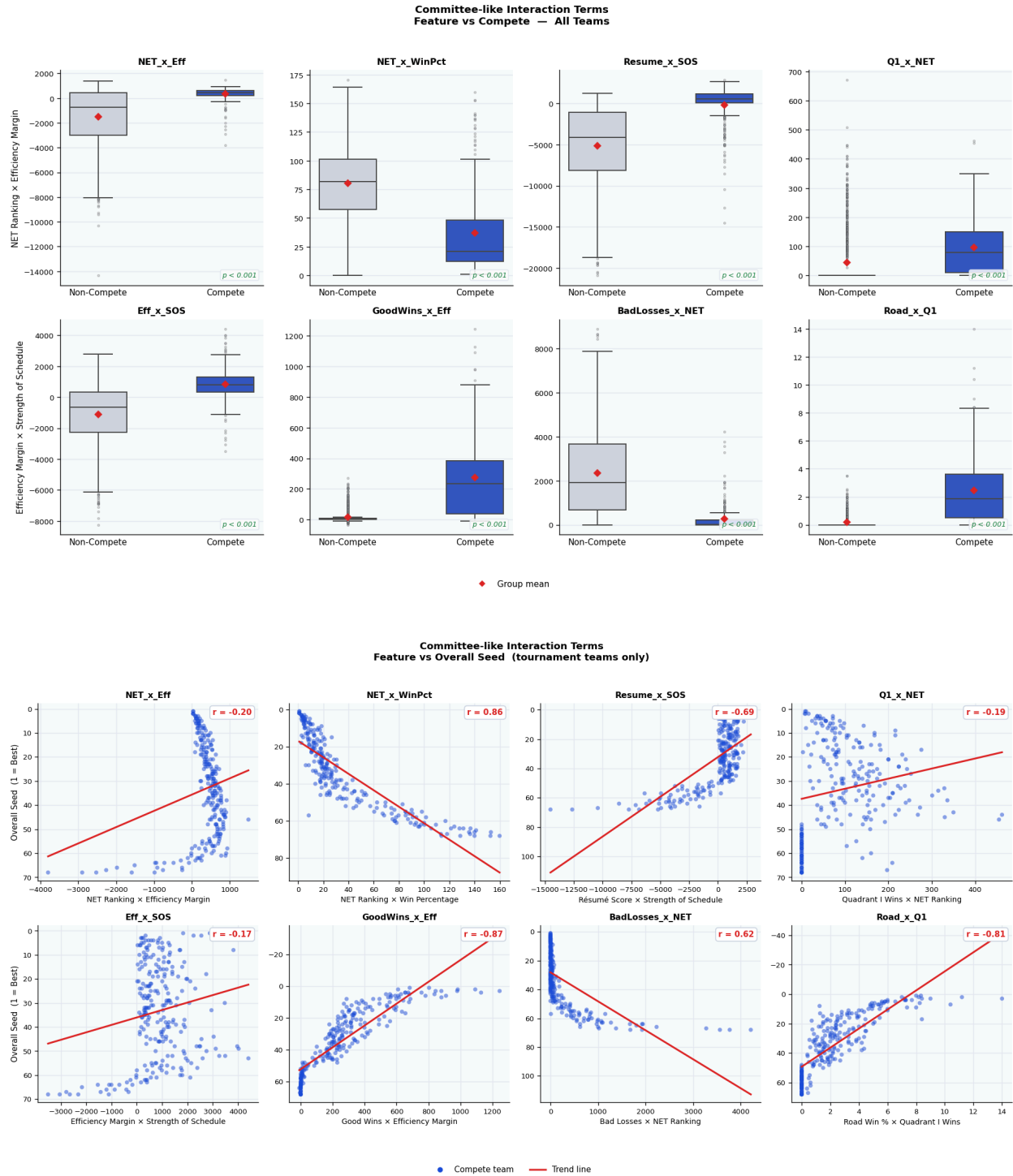


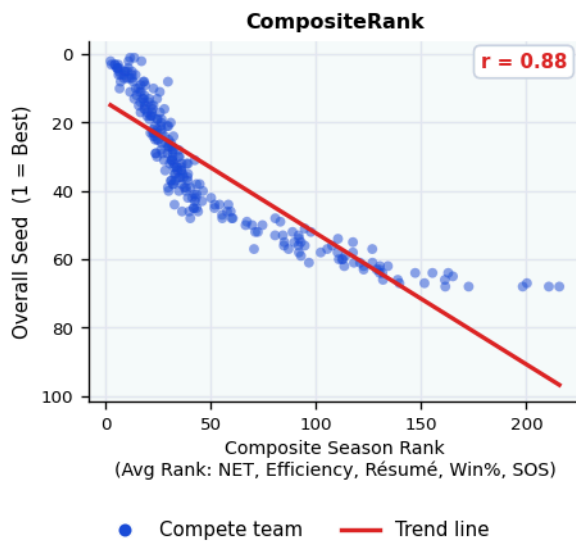
Figure 11: Average Composed Features on Tournament Participation and Tournament Seed

Feature Explanation:

- NET_x_Eff combines Ranking and Efficiency
- NET_x_WinPct favors both Ranking and Win Percentage
- Resume_x_SOS combines Résumé Quality and Schedule Strength
- Q1_x_NET composes Quadrant1 Wins and Ranking
- Eff_x_SOS combines Ranking and Schedule Strength
- GoodWins_x_Eff rewards teams that both beat strong opponents and dominate efficiently
- BadLosses_x_NET multiplies Harmful Losses and Ranking
- Road_x_Q1 combines Road Wins and Quadrant 1 Wins

7. Season Relative Rank Feature

**Season Relative Rank
Feature vs Overall Seed (tournament teams only)**



**Season Relative Rank
Feature vs Compete — All Teams**

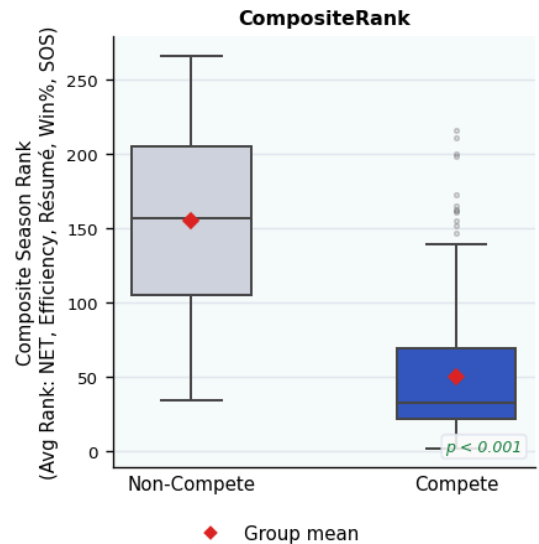


Figure 12: Average Season Relative Rank Features on Tournament Participation and Tournament Seed

Feature Explanation:

- CompositeRank summarizes season-relative standing across major dimensions like NET, efficiency, résumé, win percentage, and schedule strength.

8. Pseudo-Elo Features

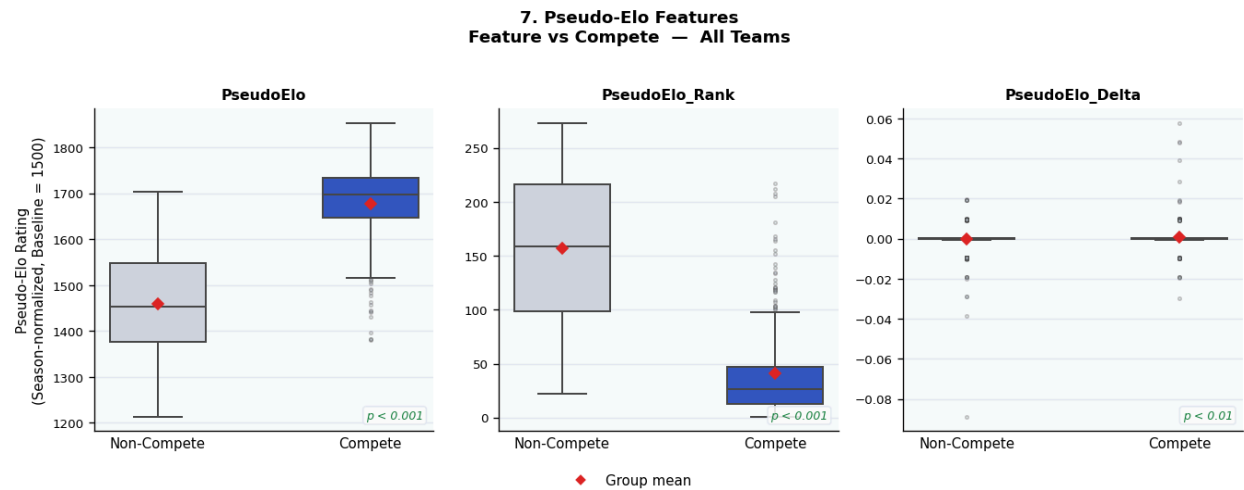
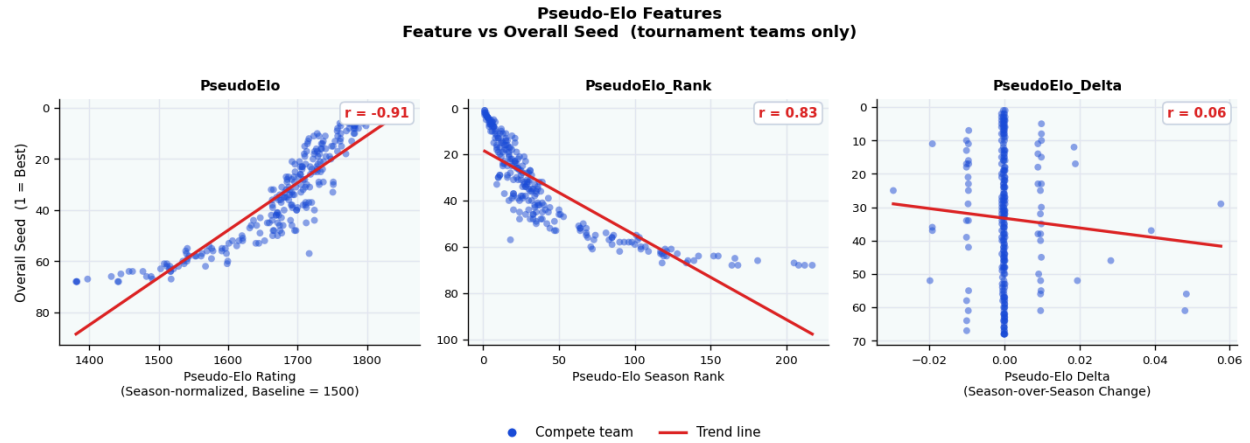


Figure 13: Average Pseudo-Elo Features on Tournament Participation and Tournament Seed

Feature Explanation:

Traditional Elo ratings update team strength through game-by-game results. Since we were working primarily with season-level features, we created Elo-like proxies that summarize team strength, relative standing, and movement over time.

- PseudoElo acts like a strength score.
- PseudoElo_Rank measures where that score stands within the season.
- PseudoElo_Delta captures whether a team's strength changed over time.

9. Feature Selection

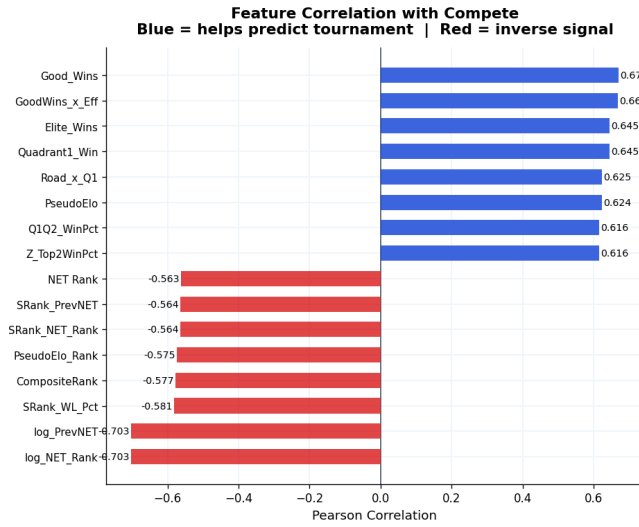


Figure 14: Top Correlated Features to Tournament Participation

To select Feature for the Classifier, we first performed correlation-based feature screening and kept features with an absolute correlation above 0.30 with the target Compete. Correlation measures the strength and direction of a linear relationship between a feature and the target. A positive correlation means higher values are associated with tournament teams and a negative correlation means lower values are associated with tournament teams. Lastly, We did not treat multicollinearity as a major issue in the classifier stage because we used tree-based models.

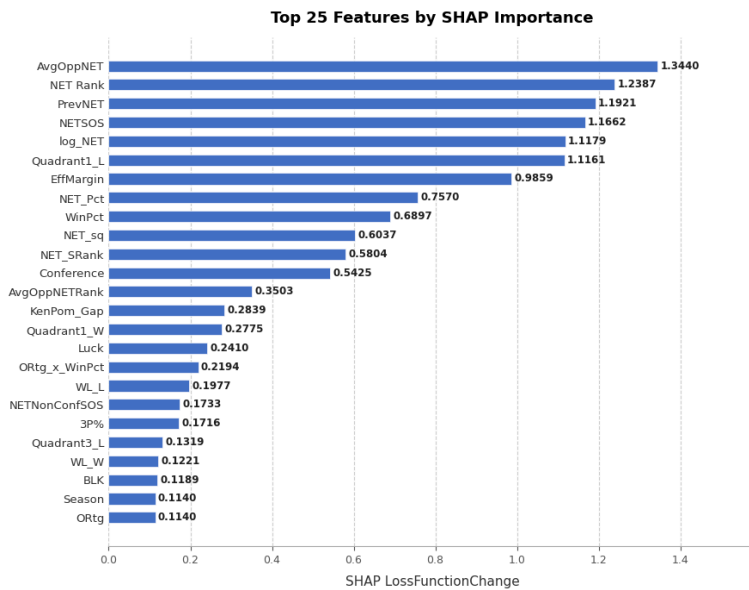


Figure 15: Top 25 Feature Importance selected by SHAP

To reduce the feature set, we used SHAP importance and kept the top 25 features. SHAP stands for SHapley Additive exPlanations. It measures how much each feature contributes to a model's predictions, averaged across observations. This is encouraging because those are exactly the kinds of variables we would expect to drive seeding.

Citations

- Chai, T., & Draxler, R. R. (2014). Root mean square error (RMSE) or mean absolute error (MAE)? – Arguments against avoiding RMSE in the literature. *Geoscientific Model Development*, 7(3), 1247–1250. <https://doi.org/10.5194/gmd-7-1247-2014>
- Chawla, N. V., Bowyer, K. W., Hall, L. O., & Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357. <https://doi.org/10.1613/jair.953>
- Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (pp. 785–794). ACM. <https://doi.org/10.1145/2939672.2939785>
- ESPN. (n.d.). Men's college basketball statistics. ESPN. <https://www.espn.com/mens-college-basketball/stats>
- Fawcett, T. (2006). An introduction to ROC analysis. *Pattern Recognition Letters*, 27(8), 861–874. <https://doi.org/10.1016/j.patrec.2005.10.010>
- Goutte, C., & Gaussier, E. (2005). A probabilistic interpretation of precision, recall and F-score, with implication for evaluation. In *Advances in Information Retrieval* (pp. 345–359). Springer. https://doi.org/10.1007/978-3-540-31865-1_25
- Kohavi, R. (1995). A study of cross-validation and bootstrap for accuracy estimation and model selection. In Proceedings of the 14th International Joint Conference on Artificial Intelligence (Vol. 2, pp. 1137–1143). Morgan Kaufmann. <https://www.ijcai.org/Proceedings/95-2/Papers/016.pdf>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, 30 (pp. 4766–4777). Curran Associates, Inc. <https://proceedings.neurips.cc/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf>
- NCAA. (2025, March 3). How do NET rankings work in NCAA tournament selection? NCAA.org. <https://www.ncaa.org/news/2025/3/3/media-center-how-do-net-rankings-work-in-ncaa-tournament-selection.aspx>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in

Python. Journal of Machine Learning Research, 12, 2825–2830.

<https://jmlr.org/papers/v12/pedregosa11a.html>

Pomeroy, K. (n.d.). Pomeroy college basketball ratings. KenPom. <https://kenpom.com/>

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. In Advances in Neural Information Processing Systems, 31 (pp. 6639–6649). Curran Associates, Inc.

<https://proceedings.neurips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>